

A Bayesian formulation of SCOUT's per-gene test for adaptive expression evolution

sparseou technical note — specification for `stan/scout.stan`

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Abstract

SCOUT fits three hypotheses to each gene on a single-cell lineage tree — neutral drift, stabilising selection on one optimum, and adaptation to several regime-specific optima — and reports the minimum-AICc winner as a single word. At small tree sizes that word is correct only slightly more often than chance, and it carries no statement of confidence. This note specifies the same three hypotheses as a Bayesian model comparison. The discrete model indicator is marginalised analytically, so a single fit per gene returns posterior model probabilities together with posteriors for the selection strength, the drift rate and the optima. The formulation is deliberately *not* hierarchical: genes are treated independently, exactly as SCOUT treats them, so that it can serve as an honest baseline against which pooling across genes is later measured.

1 What changes, and what does not

Given a rooted lineage tree, a gene's expression at the tips, and a *supplied* partition of the tips into regimes, SCOUT fits three models by maximum likelihood, penalises each by its parameter count through AICc, and returns the argmax. Two properties of that procedure motivate what follows.

First, the output is a hard label. A gene whose evidence is evenly split between two hypotheses receives the same kind of answer as one whose evidence is overwhelming. Second, AICc is an asymptotic criterion, and it is being applied at tree sizes as small as $n = 32$.

Replacing the argmax with a posterior addresses both: integrating over parameters rather than maximising, and reporting $\Pr(M \mid \mathbf{y})$ rather than $\arg \max$. What does *not* change is the scientific question, the likelihood, or the requirement that the adaptive landscape be supplied.

2 Setup and notation

Let the tree be rooted and **ultrametric**, with n tips and height H . Fix one gene and let $\mathbf{y} \in \mathbb{R}^n$ be its expression across the tips, standardised to zero mean and unit variance. Write

Symbol	Meaning
K	number of supplied regimes, $k = 1, \dots, K$
$r(u) \in \{1, \dots, K\}$	regime painted on the branch at time u — data, not inferred
H	root-to-tip time, identical for every tip
$D_{ij} = 2(H - t_{\text{mrca}(i,j)})$	patristic distance between tips i and j
$C_{ij} = t_{\text{mrca}(i,j)}$	shared root-to-MRCA path length

Throughout, $\mathbf{K}(\alpha)$ denotes the $n \times n$ matrix with entries

$$[\mathbf{K}(\alpha)]_{ij} = e^{-\alpha D_{ij}}, \quad (1)$$

which is an elementwise exponential of $-\alpha \mathbf{D}$, not a matrix exponential. Both Ornstein–Uhlenbeck hypotheses share the covariance

$$\boldsymbol{\Sigma}(\alpha, s, \tau) = s^2 \mathbf{K}(\alpha) + \tau^2 \mathbf{I}. \quad (2)$$

3 The three hypotheses

A single discrete parameter carries the scientific question,

$$M \in \{\text{BM1}, \text{OU1}, \text{OUx}\}, \quad \Pr(M = m) = \pi_m, \quad \pi = (\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}), \quad (3)$$

and everything else in the model exists to give it something to weigh. All three hypotheses are multivariate normal over the tips; they differ only in the *mean* — whether there is an optimum at all, and whether it is the same everywhere on the tree.

Neutral drift. Brownian motion started at θ_{bm} , with no mean reversion, so variance accumulates with shared ancestry:

$$\mathbf{y} \mid M = \text{BM1} \sim \mathcal{N}_n(\theta_{\text{bm}} \mathbf{1}_n, \sigma_{\text{bm}}^2 \mathbf{C} + \tau_1^2 \mathbf{I}). \quad (4)$$

One global optimum. An Ornstein–Uhlenbeck process

$$dX(t) = \alpha(\theta - X(t)) dt + \sigma dW(t), \quad (5)$$

with the root drawn from its stationary distribution about that optimum. Being stationary, the mean is θ everywhere and the covariance depends on the tips only through patristic distance:

$$\mathbf{y} \mid M = \text{OU1} \sim \mathcal{N}_n(\theta_1 \mathbf{1}_n, \boldsymbol{\Sigma}(\alpha_1, s_1, \tau_2)). \quad (6)$$

The scale is parameterised by the *stationary* standard deviation $s^2 = \sigma^2/(2\alpha)$ rather than by σ , since s is comparable across values of α and takes a prior on the same footing as the standardised data; $\sigma = s\sqrt{2\alpha}$ is recovered afterwards.

One optimum per regime. The same process, but with an optimum that is piecewise constant along the tree, $\theta(u) = \theta_{r(u)}$:

$$\mathbf{y} \mid M = \text{OUx} \sim \mathcal{N}_n(\mathbf{W}(\alpha_3) \boldsymbol{\theta}, \boldsymbol{\Sigma}(\alpha_3, s_3, \tau_3)), \quad \boldsymbol{\theta} \in \mathbb{R}^K. \quad (7)$$

Nesting. BM1 is the $\alpha \rightarrow 0$ limit of OU1, and OU1 is OUx with all θ_k equal. They are nonetheless fitted as three hypotheses with their own parameters rather than as restrictions of one, so that each carries an uncontaminated marginal likelihood.

4 The exposure matrix

Integrating (5) along the root-to-tip path of tip i gives

$$X(H) = e^{-\alpha H} X(0) + \int_0^H \alpha e^{-\alpha(H-u)} \theta(u) du + \sigma \int_0^H e^{-\alpha(H-u)} dW(u). \quad (8)$$

Let $\mathcal{P}(i)$ be the decomposition of that path into maximal segments (k, a, b) , meaning regime k occupies the time interval $[a, b]$. With the root started at its own optimum, the stochastic term has mean zero and

$$\mathbb{E}[y_i] = e^{-\alpha H} \theta_{r(0)} + \sum_{(k,a,b) \in \mathcal{P}(i)} \theta_k \left(e^{-\alpha(H-b)} - e^{-\alpha(H-a)} \right) = [\mathbf{W}(\alpha) \boldsymbol{\theta}]_i. \quad (9)$$

Each row of $\mathbf{W}$ records how much of that tip's ancestry sits in each regime, discounted by how quickly the process forgets: a tip that entered a regime long ago has arrived at its optimum, while one that entered recently still carries its ancestor's.

Proposition 1 ($\mathbf{W}$ is row-stochastic). *For every tip i and every $\alpha > 0$, $\sum_k W_{ik}(\alpha) = 1$.*

Proof. The segments of $\mathcal{P}(i)$ tile $[0, H]$ contiguously, so the sum in (9) telescopes to $e^{-\alpha(H-H)} - e^{-\alpha(H-0)} = 1 - e^{-\alpha H}$. Adding the root term $e^{-\alpha H}$ gives unity. $\square$

The tip mean is therefore a genuine weighted average of the optima. This is worth verifying numerically rather than assuming: it fails immediately if the regime painting or the segment decomposition is wrong, and it is the cheapest available check on both.

5 Priors

Parameter	Prior	Role
$\theta_{\text{bm}}, \theta_1, \theta_k$	$\mathcal{N}(0, s_\theta^2)$	optima — see §7
$\sigma_{\text{bm}}, s_1, s_3$	$\mathcal{N}^+(0, s_{\text{scale}}^2)$	drift rate / stationary spread
h_1, h_3	$\text{LogNormal}(0, s_h^2)$	phylogenetic half-life, in tree heights
τ_1, τ_2, τ_3	$\mathcal{N}^+(0, s_\tau^2)$	measurement error

Selection strength is parameterised by half-life relative to tree height rather than by α directly,

$$\alpha_m = \frac{\ln 2}{h_m H}, \quad m \in \{\text{OU1}, \text{OUx}\}, \quad (10)$$

so that $h = 1$ reads as “the optimum is reached over the tree's lifetime”. The reparameterisation keeps the sampler away from both degenerate ends: $\alpha \rightarrow 0$, where the process is indistinguishable from drift, and $\alpha \rightarrow \infty$, where it is indistinguishable from white noise.

6 Inference

Stan cannot sample a discrete parameter, so the indicator is summed out. Writing ϕ for the collection of continuous parameters and $\ell_m(\phi) = \log p(\mathbf{y} \mid \phi, M = m)$, the sampled log-density is

$$\log p(\mathbf{y}, \phi) = \log \sum_m \exp\{\log \pi_m + \ell_m(\phi)\} + \log p(\phi). \quad (11)$$

Because $\Pr(M = m \mid \mathbf{y}) = \int \Pr(M = m \mid \phi, \mathbf{y}) p(\phi \mid \mathbf{y}) d\phi$, the model probability is the posterior mean of a quantity available at every draw. No bridge sampling and no separate marginal-likelihood machinery is required:

$$\Pr(M = m \mid \mathbf{y}) = \mathbb{E}_{\phi \mid \mathbf{y}} \left[\frac{\pi_m \exp \ell_m(\phi)}{\sum_j \pi_j \exp \ell_j(\phi)} \right]. \quad (12)$$

This is a Rao–Blackwellised estimator: the conditional probability is evaluated exactly at each draw and averaged, rather than the indicator being sampled and its visits counted.

The output per gene is three numbers summing to one. A gene whose evidence is genuinely ambiguous returns something near (0.3, 0.4, 0.3) instead of a confident single word — which is the entire point of the exercise.

7 The prior on the optima is the Occam penalty

Of all the priors in the model, s_θ is the one that is not a nuisance. Marginalising the optima out of (7) analytically, which is possible in closed form because the mean is linear in θ , gives

$$p(\mathbf{y} \mid M = \text{OUx}, \alpha, s, \tau) = \mathcal{N}_n(\mathbf{y} \mid \mathbf{0}, \Sigma(\alpha, s, \tau) + s_\theta^2 \mathbf{W}(\alpha) \mathbf{W}(\alpha)^\top). \quad (13)$$

The penalty is now explicit. Freeing K optima inflates the covariance by $s_\theta^2 \mathbf{W} \mathbf{W}^\top$, a rank- K term, and the resulting log-determinant grows like $K \log s_\theta$. That is what replaces AICc’s $2k$: a larger s_θ improves the fit but is charged for the additional freedom. Set s_θ wide and OUx can never win; set it narrow and it wins too easily.

Nothing else in the model carries comparable leverage over the answer. It is therefore exposed as data rather than fixed internally, and a sweep across it belongs in the results rather than in a robustness appendix. Note that (13) also suggests a more efficient implementation than the one specified here, in which θ is never sampled; that is deliberately deferred, since posteriors for the individual optima are wanted as an output.

8 Numerical considerations

- A fixed $10^{-8} \mathbf{I}$ is added to every covariance. As $\alpha \rightarrow 0$, $\mathbf{K}(\alpha) \rightarrow \mathbf{11}^\top$, which has rank one; without the jitter every Cholesky factorisation fails at the low- α end of the prior.
- h_m is bounded to $[10^{-4}, 10^4]$, keeping $\mathbf{K}(\alpha)$ clear of overflow and underflow.
- BM1 is evaluated in the eigenbasis of $\mathbf{C}$. Since $\mathbf{C}$ and $\mathbf{y}$ are both data, the decomposition and the rotation of $\mathbf{y}$ are performed once, leaving only scalars to vary — so BM1 costs $O(n)$ per gradient evaluation rather than $O(n^3)$. The two OU hypotheses retain independent α , as OUwie does, and so require a Cholesky factorisation each.
- Chains are started from *jittered* initial values. Identical starts leave $\hat{R}$ unable to detect between-chain disagreement: a 256-tip fit reported $\hat{R} = 1.09$ from identical inits and $\hat{R} = 1.01$ once jittered.

9 Scope, and deviations from SCOUT

Deliberately excluded: any sharing of information across genes — that is the hierarchical model this specification exists to be a baseline for. Regimes are supplied, never inferred. Expression is Gaussian

on log-normalised counts, with no count model. Trees are ultrametric, and the dense Cholesky caps practical tree size near $n = 256$.

Three deviations from SCOUT are genuine and should be treated as open until checked against the original OUwie implementation.

- **The measurement-error term τ .** SCOUT's smoothed arm has no such parameter, though their EM and tip-fog arms do. It is included here because lineage smoothing multiplies the data by a similarity matrix $\mathbf{S}$, giving a covariance $\mathbf{S}\mathbf{S}^\top$ that no Ornstein–Uhlenbeck process can produce. A generative model fitted to smoothed data absorbs that mismatch as spurious signal.
- **Regime painting.** SCOUT paints internal branches using `ape::ace` under an equal-rates Mk model. The present implementation uses a deterministic parsimony rule, which coincides on simulated trees where clades are regime-coherent, and may not on real data.
- **Root treatment.** The root is placed at its own optimum here. OUwie exposes a `root.station` option; if their setting differs, the OU1 and OUx likelihoods differ in a way that no amount of correctness in the Stan program would reveal.

10 Recovery study I: neutral drift

Before any model comparison is attempted, each hypothesis should be checked in isolation: simulate a gene from it, fit that hypothesis alone, and ask whether the generating parameters come back. This section does so for BM1. The gene is simulated from exactly the model being fitted, BM1 is fitted alone (no competing hypotheses, no model indicator), and the data are deliberately *not* standardised, so σ , τ and θ retain their generating units and recovery is a real check rather than a check up to an unknown rescaling.

What is being asked

The generating process is Brownian motion on a height-one ultrametric tree, observed with additive noise:

$$y_i = X(t_i) + \varepsilon_i, \quad X \sim \text{BM}(\theta_{\text{bm}}, \sigma_{\text{bm}}^2), \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau^2), \quad (14)$$

with $\theta_{\text{bm}} = 2$, $\sigma_{\text{bm}} = 1$, $\tau = 0.3$. The three parameters play quite different roles, and it is worth being explicit about what success would even look like for each.

σ_{bm} , **the drift rate.** Estimated from the $\binom{n}{2}$ pairwise contrasts, whose expected squared differences grow with patristic distance. Information accumulates with n ; this should recover sharply.

τ , **the measurement error.** Identified by *sister pairs*. Two tips separated by a very short path should be nearly identical under pure drift; the extent to which they are not is τ . A tree with many recent splits is informative about τ ; a star tree would not be.

θ_{bm} , **the root state.** This is the interesting one, and the reason the study is worth running. Every tip descends from one root, so the n tips do not furnish n independent observations of θ . The generalised-least-squares variance is

$$\text{Var}(\hat{\theta}_{\text{bm}}) = (\mathbf{1}^\top \mathbf{V}^{-1} \mathbf{1})^{-1}, \quad \mathbf{V} = \sigma_{\text{bm}}^2 \mathbf{C} + \tau^2 \mathbf{I}, \quad (15)$$

which is computable before any sampling and is far larger than the naïve $\sqrt{\text{var}/n}$. Writing $n_{\text{eff}} = (\sigma^2 + \tau^2)/\text{Var}(\hat{\theta})$ for the number of independent observations the tree is effectively worth:

n tips	$\text{sd}(\hat{\theta}_{\text{bm}})$ from (15)	if tips were iid	n_{eff}
32	0.443	0.185	5.6
64	0.407	0.131	6.6
128	0.377	0.092	7.7
256	0.353	0.065	8.7
512	0.333	0.046	9.8

Sixteen-fold more data buys roughly four more effective observations: n_{eff} grows like $\log_2 n$, not like n . This is not a defect of the estimator but a property of the design — a phylogeny is a deeply nested sampling scheme, and the root is the one quantity all of it shares. It gives a sharp, pre-registered prediction: **a correct implementation must return a posterior for θ_{bm} roughly four times wider than iid intuition suggests, and that width must barely shrink as n grows.** An implementation that reported $\text{sd} \approx 0.09$ at $n = 128$ would be confidently wrong, and no amount of posterior-predictive checking on the tips would reveal it.

What Stan returns

BM1 was fitted alone to one simulated dataset at each of four tree sizes, four chains, 1000 warm-up and 1000 sampling iterations. Every fit reported $\hat{R} \leq 1.005$ and zero divergent transitions, and each took under 1.3 seconds.

n	θ_{bm} (truth 2.0)	σ_{bm} (truth 1.0)	τ (truth 0.3)	$\text{sd}(\theta)$ predicted	$\hat{R}$
32	1.89 ± 0.28	0.55 ± 0.27	0.50 ± 0.16	0.443	1.003
64	2.65 ± 0.45	1.09 ± 0.16	0.27 ± 0.15	0.407	1.002
128	2.41 ± 0.376	1.008 ± 0.114	0.23 ± 0.11	0.377	1.005
256	1.77 ± 0.40	1.15 ± 0.08	0.16 ± 0.09	0.353	1.004

Two things in that table matter more than the fact that the intervals contain the truth.

First, at $n = 128$ the posterior standard deviation of θ_{bm} is 0.3756 against the 0.377 predicted by (15) — agreement to 0.4% on a quantity computed before sampling began. That is a far stronger check than coverage of a point value, because it tests the covariance structure rather than just the location.

Second, the $\text{sd}(\theta)$ column is effectively flat across an eightfold increase in tips while the iid expectation falls from 0.185 to 0.065. The predicted non-shrinkage is visible in the fitted output, not merely in the algebra.

At $n = 32$ the fit is genuinely poor: $\sigma = 0.55$ against a true 1.0, with τ correspondingly inflated to 0.50. This is the drift–noise trade-off, not a bug. Both parameters add variance, and 32 tips do not contain enough short branches to separate a slow diffusion from a noisy observation of a fast one. Across 150 replicate datasets the posterior means of σ_{bm} and τ correlate at $r = -0.73$.

Calibration

Recovery on a single dataset cannot distinguish a correct implementation from a lucky one. The model was therefore refitted to 150 independently simulated datasets at $n = 128$, asking how often the central

credible intervals contain the generating value. Nominal coverage should be recovered within Monte Carlo error, and is:

Parameter	50% interval	90% interval	mean posterior mean
θ_{bm}	0.467 ± 0.080	0.853 ± 0.057	1.935 (truth 2.0)
σ_{bm}	0.520 ± 0.080	0.927 ± 0.042	1.007 (truth 1.0)
τ	0.520 ± 0.080	0.873 ± 0.053	0.265 (truth 0.3)

All six figures sit within two standard errors of nominal, and no fit in the 150 produced a single divergent transition. The one visible bias is in τ , whose posterior mean averages 0.265 against a true 0.3: the half-normal prior pulls a small positive parameter toward zero. Interval coverage is unaffected, so this is prior shrinkage behaving as specified rather than an error.

Scope of this check. Coverage is evaluated at one fixed parameter value, not averaged over the prior, so it is a frequentist calibration statement rather than simulation-based calibration. Full SBC — drawing the generating parameters from the prior at each replicate — is the stronger test and is now cheap, since a fit takes well under a second.

11 Validation status

Check	Status	Result
$\mathbf{W}(\alpha)$ vs. reference construction	done	agrees to 10^{-16} ; rows sum to 1
BM1 likelihood vs. independent ML	done	exact to four decimals
OU1, OUx vs. independent ML	done	agree to 0.005 where the optimiser converges
Sampling behaviour at $n = 256$	done	$\hat{R} \leq 1.02$, ESS ≥ 250 , jittered inits
Independent ML vs. original OUwie	<i>open</i>	gates everything below
Accuracy and calibration on truth-labelled data	<i>open</i>	
Behaviour across tree size	<i>open</i>	

The first four checks establish that the program computes the intended likelihood. They say nothing about whether that likelihood is SCOUT's, which is the purpose of the fifth — and which matters, because a faithful reimplement of a drifted reimplement is still drifted.